

Variable and Method Flow

Data logic, variable movement and model workflow across all Day 12 notebooks

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Purpose

This document explains not only the variable names, but also the logic of how data moves through the analytical workflow. The flow is written in plain language so every step has a reason.

Flow area	Input	Transformation / method logic	Output / use
Notebook 1 - Advertising regression	TV, Radio, Newspaper, Sales	Split features and target. Fit simple TV-only regression, then fit multiple regression with all channels. Compare metrics and residuals.	Sales predictions, coefficients, R2, adjusted R2, residual checks and business driver ranking.
Notebook 2 - Regularized regression	Scaled numeric regression features	Fit Linear Regression as baseline. Apply Ridge/Lasso/ElasticNet across alpha values. Use Lasso path and LassoCV to understand feature shrinkage.	RMSE curves, best regularized model, selected alpha, zero coefficients and VIF risk table.
Notebook 3 - Count models	Bike Sharing predictors and cnt target	Avoid leakage from casual and registered. Use time-aware split. Compare linear baseline with Poisson and Negative Binomial count models.	MAE, Mean Poisson Deviance, overdispersion evidence, AIC/BIC and model recommendation.
Notebook 4 - Logistic Regression	Customer campaign features and response target	Preprocess numeric/categorical features. Fit Logistic Regression. Convert probabilities into class labels using thresholds.	Confusion matrix, ROC-AUC, PR-AUC, threshold tradeoff, calibration and lift chart.
Notebook 5 - Diagnostics	Same dataset families	Show coefficient-solving logic, interval logic, residual assumption checks, influence diagnostics and advanced model variants.	Missing-topic closure and deeper explanation layer.

Core variable logic

Variable group	Plain-language logic
X variables / features	The information available before prediction. Example: TV budget, Radio budget, bike weather variables or customer behaviour signals.
y variable / target	The value the model tries to predict. Example: Sales, cnt rental count or campaign response.
X_train and y_train	The part of data used to learn coefficients. The model is allowed to see these rows while fitting.
X_test and y_test	The unseen holdout data used to check whether the model generalizes. It is not used for learning coefficients.
Predictions	The model output for test rows. Regression gives numeric predictions; logistic regression gives probabilities and class labels.
Residuals	Actual minus predicted. Residuals show what the model missed.
Metrics	Numbers that summarize quality, such as RMSE, MAE, R2, ROC-AUC or PR-AUC.

Logic chain

From data to decision

1. Understand target type. 2. Select the right model family. 3. Split data safely. 4. Preprocess without leakage. 5. Fit model on training data. 6. Predict on test data. 7. Evaluate with target-appropriate metrics. 8. Diagnose assumptions and risks. 9. Translate output into a business recommendation.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.